PROJECT DEFINITION REPORT

QUALITY ASSURANCE PRE-WARNING SYSTEM

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1.0 Introduction

Large enterprises have the luxury of being able to afford complex, customized computer-vision solutions with multiple types of sensors, varying stations along a production line and can better afford OEM installation and on-going support costs. The high initial capital investment cost is less of a limiting factor for large enterprises compared to SMEs. There are several drivers behind this high price tag including: OEM deployment costs including installation staff and lengthy employee training regiments, price premiums on OEM components including servers, sensors and cameras, prohibitively complex closed proprietary software, and customization needed to fit existing equipment and trained models to fit the live production environment (Rosenbrock, 2017).

Quality control and inspection can occur at every stage of the manufacturing process from the receiving of raw materials, to individual manufacturing steps, to final inspection and packaging (Keen, 2021). At each stage, workers are responsible for detecting cosmetic, physical and material defects. Studies show poor inspection quality is accountable for as much as 20% of production failures at some facilities (Lewis, 2016). Industry 4.0 addresses quality control issues by standardizing, automating and improving quality production while increasing communication between stakeholders and transparency. Labor shortages have made quality assurance (QA) workers harder to find and afford while existing solutions are cost prohibitive for SMEs. Inadequate QA inspection runs the risk of high defective rates, client returns and reputational damage, and inflicts the expense and emissions of the return process.

Consequently, driven by labor shortages, SMEs have a need to augment their QA with non-laborious methods like computer vision inspection systems, without large capital investments.

The proposed computer-vision system will require an interdisciplinary engineering approach. Electrical engineering expertise will be needed in the wiring of auxiliary components such as cameras, lidar and other data collection equipment to a server. Additionally, a working model or showcase of the system will be constructed, where wiring and setup of DC motors for a conveyor belt will be needed. Mechanical engineering expertise will be utilized for the CAD design and manufacture of housing for the camera and other components such as mounting brackets, internal supports and ventilation. Manufacturing/software engineering expertise will play an integral role with the development of the code behind the computer-vision system.

The computer-vision system will be powered by a deep learning architecture. Out of the 3 major categories of neural networks; artificial neural network (ANN), recurrent neural network (RNN) and convolution neural network (CNN), CNN was chosen as the main architecture. CNN was selected due to its strengths and scalability in being able to automatically detect the presence of visual elements in images via classification. Classification can reach high accuracy in the presence of sufficient training data. Classification allows a system to recognize the presence of defects and alert human counterparts. CNN architectures can be customized in scale, by adding or removing, convolutional network layers, randomization, dropouts and other regularization

techniques to balance speed versus accuracy within a SMEs available hardware (Rosenbrock, 2017).

As a result, the computer-vision system powered by the deep learning CNN algorithm will then trigger a pre-warning system, such as visual and audio cues that will alert factory workers of a potential product defect. The system should be able to match or exceed, in speed or accuracy, human performance in manufacturing defect detection.

2.0 Needs and Constraints Identification

A way to design a cost-effective deep learning computer vision system (DLCVS) for small to medium size enterprises (SMEs), that will lead to more sustainable operations and resolve labor shortages during the covid-19 pandemic. The computer vision system is designed around Region-Based Convolutional Neural Networks (R-CNN), to be robust and dynamic enough to be used in a variety of facilities. The intention of this design is to be affordable for SMEs, provide robust performance while being easy to train and use, while addressing labor shortages and meeting industry quality standards requiring visual inspection like ISO 9000 QMS.

2.1 Description/overview of company and stakeholders

This solution was peer-proposed by group leader Wesley Burchnall who expressed his desire to assist SMEs (primary stakeholders) meet industry quality standards. Pointing out that the computer vision system we are developing will help Kelowna's orchards and produce sector, who have especially experienced labor shortages. With the city's unemployment rate dropping below BC's average of 5.9 percent to 4.4 percent, the orchards and produce sector has been reeling as they struggle to attract workers (secondary stakeholders).

By implementing our DLCVS, SMEs make customers into tertiary stakeholders who will benefit from enhanced product quality. They may also benefit from cheaper price points due to lower cost of manufacturing from reduced labor costs. A review of a Global Manufacturing Company's facility published in the Harvard Business Review, found that an inadequate QAS help contribute to the facility, ' ...sending out items, 15% of which (on the average) have one or more major defects'. A reduction in defective products being supplied by SMEs, will also directly impact resources consumed by remediation; leading to less greenhouse gases being released into the environment, another secondary stakeholder.

Finally, Dr. Hassan Iqbal, the faculty advisor has been guiding the project through project management advice and strategies to enhance the quality of the project, team cohesion and output.

2.2 Description of previous solutions and opportunity

When reviewing the existing solution landscape, the solutions we found were all targeted at large enterprises and none were focused on meeting the needs of SMEs. Additionally, over 80% of the solutions were single-industry solutions targeting a fixed market segment in one industry. The few multiple-industry solutions were impressive in capability yet prohibitively expensive. Appendix 2 provides more details.

Our opportunity is aimed at leveraging artificial intelligence (AI) and computer vision (CV) in our DLCVS, which will be an improvement on the systems of yesterday. As the DLCVS can detect the imperfections as little as 0.01 mm, at distances of 10cm or 1cm at distances of 5 feet, DLCVS can alert operators in real time and employ Electrical Actuators Based Conveyor belts (EABCBs) for sorting identified products (Prikhodko, 2021).

1. Manual Inspections - This method of quality control is done solely on human observation and no inspection devices or technology according to Creaform, C.
2. First-article inspections (FAIs) - This method according to Creaform, C., ‘does depend on machinery and may seem to be fully automated, but does depend on intervention by an experienced CMM specialist or metrologist to manually load, operate and unload the CMM’.
3. The Review Scanning Electron Microscope (Reference SEM) is fully automated, and consists of two main categories: Die to Die (D2D) detects hotspots by comparing SEM images with the reference SEM image and classifies them according to the rules already stored on the server and Die to Database (D2DB) inspection which finds defects by comparing SEM images with the reference layout (or target pattern) . According to J. Kim *et al*.,” The main disadvantage of D2D inspection is that we need to select a reference SEM image for each pattern layout, which is hardly automated, while in the D2DB it is difficult to accurately extract the contour lines from noisy SEM images, and due to the diversity and irregularity of patterns and defects, this method is often limited in practice”
4. You Only Look Once (YOLO) is a single-stage object detection network that is capable of predicting the location of objects through bounding boxes with high detection accuracy at considerable speed.”(Deng1 ,2020). The YOLO method uses one look at the image that is provided to the input and decides whether the object is meant to carry a certain action out or not. It assigns importance weightage to the different parts of an image to differentiate between them and provide an output.
5. Reluctant Neural method: It uses a bunch of questions to narrow down to a premise " is there a different spectrum of colors" yes/no, does the particular spectrum of colors mean it's a crack "yes/no". These questions are then used to narrow down the result that is required off the Ai and when the image is inputted to the AI the questions that were used to train the AI are used to deliver an output.

2.3 Description of scope and constraints

Scope:

The aim of this project is to develop a R-CNN based DLCVS, that understands and differentiates the quality of products at various manufacturing stages for quality assurance control at price points affordable to SMEs most affected by the covid-19 pandemic.

The system consists of an input device(camera) which feeds images of products into the DLCVS for processing. The program will then cross reference the real time images versus the AI trained to recognize images containing defects. When abnormalities are detected, the DLCVS system will flag the product as defective and alert quality control staff in real time through an output medium. Mediums could include auditory warnings (for visually impaired), visual warnings in the form of a video display or warning-light (for auditory impaired) or a combination of the above.

Constraints:

1) The live-video stream must have a minimum resolution equal to or greater than the convolution layer input size. The system can apply scaling and cropping to adjust a larger video stream resolution to fit the convolution layer input size.
2) The lighting used for training should be consistent with the lighting in the live production environment. Our system uses images captured from the live training environment which should eliminate this constraint.

Figure 1: Image demonstrating effects of different lightning environments on image quality (Rosenback, 2017)

3) The live-video stream from the camera should ideally be free of visual defects and in focus. This includes obscuring from objects in the scene that would cause occlusion variation and dust particles.

Figure 2: Demonstration of effect of occlusion variation (Rosenback, 2017)

4) Temperature control and humidity levels may be a constraint. Condensation on camera lense during cold days would directly affect the ability of the system to detect defects. Thus the operating temperature should be maintained within standard norms.

Figure 3: Demonstration of effect of condensation on image quality

5) The camera may be positioned laterally, vertically above or below, so the material selected for the device casing must be able to withstand the force of gravity of the internal components across all six axes.

Internal Constraints for the project :

1) Time: The project needs to be completed and delivered by April 15, 2022
2) Cost: The team has to stay within a prototyping budget of three hundred dollars
3) Training Data: To train the AI effectively a minimum of one hundred images is needed. We will need to find the images of a certain product or take those images ourselves.

Economic consideration:

The major cost incurred would be the installation and the purchase of the parts to design the system. Once the system is set up, the recurring cost will only be the electricity consumed by the camera and the computer system.

This is an automated system which would labor in quality assurance departments and lead to higher profits for the company that uses this and higher unemployment in that sector.

Environmental consideration:

The camera used by the input could vary according to the choice of the consumer based on their needs, however most cameras consist of microchips that contain gold which can be recycled once the life of the camera is over. The battery of the camera can also be safely disposed of by going to a recycling plant.

However, implementation of this system will positively impact the environment, through the implementation of our CV system, SEMs will reduce their impact on the environment in more ways than one, such as:

1) Transportation: The transportation of goods back to its producers as well as employing less people.
2) Energy Usage: A reduction of energy expended on products that are deemed substandard before successive production stages.

Health and safety risks :

Risks associated with automation of manufacturing processes is more mental awareness than physical. As according to Lee, S. M. et al., ‘ ... the excessive adoption of automation has resulted in immoderate dependence on automation, and it has generated a failure to maintain operators’ attention in detecting and monitoring situations’. Which may cause Subjective Cognitive Decline (SCD), that according to the CDC is, ‘ ...is a form of cognitive impairment and one of the earliest noticeable symptoms of Alzheimer’s disease and related dementias’. Although physical risks are minimized, a loss of concentration while working around machinery may lead to bodily harm.

3.0 Problem Specifications

Our specifications are intended to reduce component quantities and costs, the installation and operator expertise required and minimize model-retraining time. Large OEM packages often have a series of proprietary hardware including cameras, connectors, cables, receivers and so-forth (KEYENCE, n.d.). With OEM, if a camera breaks, one is limited in the replacements one can use.

Figure 4: A selection of cameras offered as part of Cognex In-Sight 2D vision line.

3.1 Camera Requirements

Each with a moderate markup common in proprietary hardware. Removing proprietary hardware, reduces cost of entry and repair/maintenance costs benefiting SMES. Our system will work with any camera from any manufacturer provided it meets the following specifications:

Feature	Specification
Color System	RGB 8-24 Bit or RGB 8-24 Bit Depth Channel
Resolution	720p, 1080p or 4k
FPS	30, 60 or 120 frames per second
Minimum and Maximum Focus Depths	Must match distance from placement position to inspection position.

Lens Type	Wide angle, macro, telephoto/microscopic, fish-eye (less ideal)
Connection Type	USB 2.0, 3.0 or 3.1 standard, or 802.11 a/g/n

3.2 Server Requirements

Feature	Specification
Operating System	Windows/Linux
RAM	4 Gb/2Gb
Architecture	64-bit x64 Processor
Storage	650MB in addition to OS installation size and Training Image Dataset Storage
Minimum Processor Throughput	30,000 MOps/Sec
Recommended Processor Throughput	80,000 MOps/Sec
MOps per FPS (Avg)	2,567 MOps/Frame (inc Windows OS overhead)

3.3 Installation Requirements

3.3.1 Camera installation

Installation requirements vary depending on connection type. Camera minimal installation includes mounting and positioning angled. Camera must be angled and mounted within the depth of field range, as per camera OEM specifications, with respect to inspection point. Camera optional installation for wired cameras includes wiring a USB connection from the camera to the CV server. Typical SME maintenance personnel possess the expertise required to mount a camera and run cabling, so no OEM on-site visitation is required for installation.

3.3.2 Server Requirements

The software package must be installed on a pre-existing OS on a server located physically close enough to receive the WIFI signal from the camera (or relayed signal). Server installation is semi-automated with selection of installation destination, user license agreement and options for adding execution-shortcuts to desktop, start menu or start-bar for Windows users and environmental path for Linux users. Administrative privileges may be required to install new software on some machines. Most IT staff can perform these tasks, so no OEM on-site visitation is required. Remote assistance can be used to troubleshoot difficulties.

3.4 Training Requirements

3.4.1 Operators

The system pre-warns quality assurance personnel of defects coming down a line. The system runs automatically and displays warnings via a monitor with semi-live view, where the object is located within a bounding box, flashing light or audible tone. Training only requires explaining to the QA assurance personnel, what warning to be mindful of and observe.

The software is set to launch upon OS boot so turning on the computer and logging with valid network credentials will be sufficient for the program to begin operation.

3.5 Engineers and Operator Expertise Requirements

3.5.1 Data Collection:

Engineers or QA Specialists need to capture images of defects and label the image by defect type from the live production environment. This can be done real-time when the operator witnesses a defect on a production line during setup or can be simulated by leaving them stationary and capturing stills of defective items. The advantage of our system is the cameras are captured in the environment. By default, the system will also apply image randomization techniques to the dataset including random-rotations between pre-set degrees, skewing, translation and scaling.

3.5.2 Training:

1) Automatic: Application includes default training settings can be adjusted/fine tuned though optimal settings will be defaulted to. By default, training from the image dataset requires filling out a short form such as selecting image classification categories (what defects you wish to observe/alert over). Training commences automatically until stop conditions are reached. These conditions stop after a fixed number of training cycles (Epochs) or when the gap reaches a minimum threshold requiring no observation.

2) Manual: Manual training is possible also and may be added as a “stretch-goal” for our project. This would include a visual graph displaying training loss, training accuracy, test validation and test loss periodically updated throughout training.

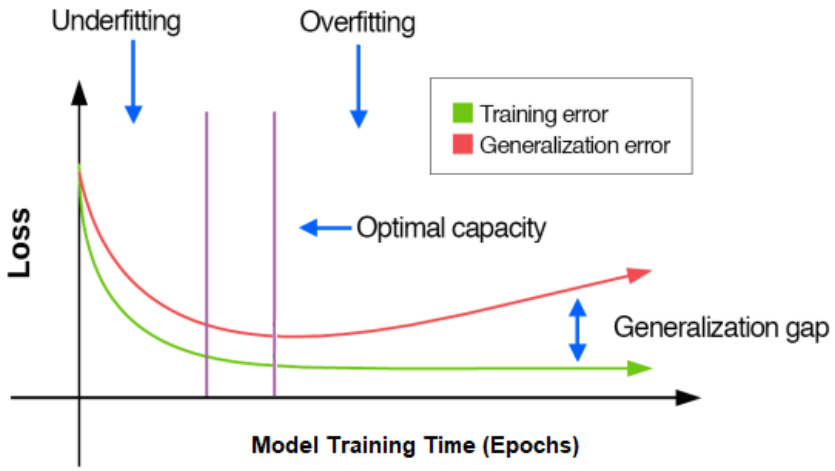

Figure 5: Illustration of Errors During Computer Vision Model Training. Modified from (Rosenbrock, 2017)

Training should be stopped when significant divergence occurs between the test-accuracy/validation and training. User expertise required is minimal and only needs the ability to interpret line graphs. Additionally, training can be set to stop after a fixed number of training cycles (Epochs) or when the gap reaches a minimum threshold requiring no observation.

4.0 Conclusion

The purpose of this report was to discuss aspects of a computer vision pre-warning quality assurance system. The scope and objectives include developing the AI software that will be responsible for powering the computer vision system, along with completing a physical prototype. We discussed how to implement the CV using region based convolutional neural network algorithms. Additional methods such as using a reluctant neural network and YOLO were also proposed. However, CNN was the final selected method due to its strengths in scalability, key element detection and high image recognition accuracy.

In regards to the physical design aspect, existing solutions including: camera orientation, electrical actuator based conveyor belts, and light source detection methods were discussed. These were later deemed ineffective due to software capabilities being able to make up for constraints like object orientation or different light environments. The next steps for the project are to implement a physical system which will include a conveyor powered by a dc motor, design of camera and auxiliary component casing, and overall showcase model for the software developed in the project.

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A2: Existing Solution Landscape

Solution Name	Multiple or Single Industry Solution (M or S)	Industry(ies) Served	Proprietary Hardware	Target Audience (SME/LE)	Cost
Clear Vision Boxes	S	Coagulated Boxes Manufacturing	Y	LE	Medium
Dark Vision Technology	S	Gas and Oil Industry - Pipe Inspection	Y	LE	High
Proof Of Concept in International Journal of Clothing Science and Technology	S	Textile and Garmet Manufacturing - Fabric Defect Dection	N/A	N/A	N/A
Proof Of Concept in The International Journal of Advanced Manufacturing Technology	S	Argicultural Industry - Bean Quality Inspection	N/A	N/A	N/A
RoboVision.ai	M	Argicultural, Healthcare, Textile Manufacturing	Y	LE	Very High
A.I. See	M	Various Manufacturing Sectors	Y	LE	Very High

Table 1: Characteristics of Existing Solution Landscape